

Ismail Kayaalp

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About

As an Electronics and Communication Engineer, I have a strong interest in robotic systems, image processing, and embedded systems. I completed internships at Turkcell and Baykar, where I gained valuable experience in both industry-leading communication technologies and advanced UAV/robotics applications. I am an enthusiastic engineer candidate, open to innovations and continuously improving my skills. With effective communication abilities, I work well within a team and contribute positively to collaborative projects.

Education

Kocaeli University, Electronic and Communication Engineering	Sept 2021 – Jun 2026
Ecole 42, computer Science	Sept 2021 – Jun 2023
Kocaeli University, English Preparatory Program	Sept 2020 – Jun 2021

Experience

TURKCELL - Software Engineering Intern Aug 2025 – Sep 2025
Completed a one-month summer internship in the Process Fixed Solutions Unit at Turkcell, where I gained hands-on experience with unit processes and internal applications, and contributed to team projects, enhancing both technical and communication skills.

BAYKAR – Long-term SWE Intern Feb 2025 – Jun 2025
Completed a long-term internship at Baykar in the Light Platforms Unit, developing software for unmanned aerial vehicles. Worked on AI-based object detection and tracking systems, focusing on their implementation and optimization on GPU platforms.
Technologies: Python, YOLO, CUDA, OpenCV, PyTorch

ERG CONTROLS - Part-time R&D Engineer Aug 2024 – Jan 2025
Working as a part-time R&D Engineer at ERG Controls, developing embedded applications for IoT devices. Focused on building and integrating IoT systems, including sensor communication and device-to-cloud data transmission.
Technologies: LoRa, BLE 5.1 , MQTT, C++ , C, I2C, SPI

ISTANBUL UNIVERSITY - Engineering Intern Jul 2024 – Aug 2024
Completed a summer internship at Istanbul University, working on a smart glasses project for visually impaired individuals. Developed object recognition and text-to-speech features, and implemented cloud-based image processing solutions.
Technologies: Python, OpenCV, YOLO, PyTorch, Cloud Systems

KOVAKTECH - Co-Founder And Software Team Member Apr 2023 – Sep 2025
Served as a founding member of KOVAK Tech contributing to the software unit focused on embedded systems image processing and data transmission Participated in the Teknofest Savaşan İHA competitions organized by Baykar in 2024 Adana and 2025 Çanakkale reaching the finals in both events Awarded Best Control And Guidance Software in 2025.
Technologies: C++ , Python, OpenCV, PyTorch, UDP

References

DR. Leyla Türker | Assoc Professor - Istanbul University | Phone: +90 532 443 48 58

Muhammed Haktan Acar | Turkcell - Senior Software Engineer | Phone: +90 533 210 05 88

Projects

Autonomous UAV Dogfight Project – Kocaeli University 2025

I developed a project enabling autonomous dogfight simulations between UAVs using YOLOv8 and CSRT tracking algorithms at Kocaeli University. I implemented object detection, tracking, and control algorithms that allow UAVs to detect and engage opponents autonomously.

Technologies: Python, PyTorch, YOLOv8, CSRT, Jetson Nano

LoRa Temperature Data Transmission Project – Kocaeli University 2023

I implemented a project for my engineering course at Kocaeli University to transmit temperature data over 2 km using LoRa technology. I designed and built dual-layer PCBs and developed firmware for STM32 microcontrollers to read and transmit sensor data.

Technologies: C++, LoRa, STM32, Altium

Temperature Control System – Kocaeli University 2022

I developed a temperature control project using MSP430 at Kocaeli University. I designed a system that activates a heating wire if the temperature drops below a set threshold and turns on a fan to cool the environment if the temperature rises above a set level. I also designed and fabricated the PCB for the system.

Technologies: MSP430, Altium, PCB Design, C